PRACTICAL 1

Project Initiation, Requirement Analysis, Sitemap, Wireframe, and GitHub Setup

1. Problem Definition

StudentHub is a web-based student portal that provides students with academic and student-related information in one place. It will include modules such as profile, academics, resources, notices, events, assignments, and attendance.

The objective is to plan and design the basic structure of the portal before actual development.

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2. User Roles

Student:

- **Register/Login**
- **View dashboard**
- **Manage profile**
- **View academics and resources**
- **View notices and events**
- **Check assignments and attendance**

Admin:

- **Manage students**
- **Add/update notices and events**
- **Manage academic resources and assignments**

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3. Main Modules / Pages

The StudentHub portal will contain at least 10 pages:

1. **Home**
2. **Register**
3. **Login**
4. **Dashboard**
5. **Profile**
6. **Academics**
7. **Resources**
8. **Notices**
9. **Events**
10. **Assignments**
11. **Attendance**
12. **Help/Contact**

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4. Functional Requirements

- **User registration and login**
- **Student profile management**
- **Academic information display**
- **Study resource access**
- **Notices and announcements**
- **Event information**
- **Assignment information**
- **Attendance information**
- **Help and contact facility**

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5. Sitemap

StudentHub

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├── Home

├── Register

├── Login

│

└── Dashboard

├── Profile

├── Academics

├── Resources

├── Notices

├── Events

├── Assignments

├── Attendance

└── Help/Contact

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6. Low-Fidelity Wireframe

The wireframe represents the basic layout of the StudentHub pages before applying final design and styling.

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7. Project Folder Structure

StudentHub/

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├── index.html

├── pages/

├── css/

│ └── style.css

├── js/

│ └── script.js

├── images/

└── README.md

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8. Key Questions

Q1. What is URL and its parts?

URL (Uniform Resource Locator) is the address used to locate a resource on the internet.

Example:

https://example.com/student/login

Parts:

- **https → Protocol**
- **example.com → Domain**
- **/student/login → Path**

Q2. How is an HTML file processed in a web browser?

HTML File

↓

Browser reads HTML

↓

HTML is parsed

↓

DOM is created

↓

CSS & JavaScript processed

↓

Webpage displayed

The browser parses HTML to create the DOM, applies CSS, executes JavaScript, and finally renders the webpage.

Q3. How is page navigation managed?

Navigation between HTML pages is managed using the <a> (anchor) tag.

Profile

When the user clicks the link, the browser opens profile.html.

Q4. How will GitHub commits be maintained?

After completing each practical or major feature, changes will be committed and pushed to GitHub.

Example:

git add .

git commit -m "Added sitemap and wireframe"

git push

Meaningful commit messages will be used to track project progress.

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9. README.md

StudentHub

StudentHub is a web-based student portal.

Features

- Login and Registration

- Dashboard

- Profile

- Academics

- Resources

- Notices

- Events

- Assignments

- Attendance

Technologies

HTML, CSS, JavaScript, Git and GitHub

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10. GitHub Repository

Repository Name: StudentHub

GitHub Link: _https://github.com/Hastee22/wt

11. Conclusion

The StudentHub project foundation was prepared by analyzing requirements, defining users and modules, creating a sitemap and wireframe, organizing the project structure, and setting up GitHub for version control.

Learning Outcome

The practical helped in understanding requirement analysis, project planning, sitemap design, wireframing, folder structuring, and GitHub version control.